

AI Plant Diagnosis System

Flask Backend | app.py | Final Year Project

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from flask import Flask, request, jsonify
import numpy as np
from PIL import Image
import pandas as pd
import os
import tensorflow as tf
tflite = tf.lite

app = Flask(__name__)

base_dir = r"C:\Users\Sri_Hari\Downloads\Farmer app\Farmer app\backend\models\tflite_models"
model_paths = [
    os.path.join(base_dir, "InceptionV3.tflite"),
    os.path.join(base_dir, "ResNet101.tflite"),
    os.path.join(base_dir, "VGG16.tflite")
]

def load_model_dynamic(path):
    interpreter = tflite.Interpreter(model_path=path)
    interpreter.allocate_tensors()
    return interpreter, interpreter.get_input_details(), interpreter.get_output_details()

meta_path = r"C:\Users\Sri_Hari\Downloads\Farmer app\Farmer app\backend\metadata.csv"
if os.path.exists(meta_path):
    df = pd.read_csv(meta_path)
    df["label"] = df["crop"] + "/" + df["disease"]
    classes = sorted(df["label"].unique())
else:
    classes = [
        "chili/bacterial_spot", "chili/healthy", "chili/leaf_curl", "chili/leaf_spot",
        "chili/whitefly", "chili/yellowish",
        "corn/common_rust", "corn/gray_leaf_spot", "corn/northern_leaf_blight",
        "corn/southern_rust", "corn/healthy",
        "potato/early_blight", "potato/late_blight", "potato/healthy",
        "sugarcane/red_rot", "sugarcane/rust", "sugarcane/yellow_leaf_virus", "sugarcane/healthy",
        "tomato/bacterial_spot", "tomato/early_blight", "tomato/late_blight", "tomato/leaf_mold",
        "tomato/target_spot", "tomato/septoria_leaf_spot", "tomato/spider_mite",
        "tomato/mosaic_virus", "tomato/yellow_leaf_curl_virus", "tomato/healthy"
    ]

DIAGNOSES = {
    "chili/bacterial_spot": "A bacterial infection causing dark, water-soaked spots on leaves and fruits. "
        "It often leads to leaf drop and fruit rot, reducing quality.",
    "chili/healthy": "The plant is healthy and growing vigorously with no visible signs of pests or diseases.",
    "chili/leaf_curl": "A viral condition causing leaves to curl, twist, and thicken. "
        "It severely stunts growth and prevents proper fruit development.",
    "chili/leaf_spot": "Fungal disease appearing as circular spots with gray centers. "
        "It causes leaves to yellow and fall off prematurely, weakening the plant.",
    "chili/whitefly": "Small insects that feed on plant sap, weakening the host. "
        "They secrete sticky honeydew promoting mold and can transmit harmful viruses.",
    "chili/yellowish": "Indicates plant stress, often due to nutrient deficiency or root issues. "
        "It results in poor growth and reduced photosynthesis.",
    "corn/common_rust": "Fungal disease forming cinnamon-brown pustules on leaves. "
        "It reduces the photosynthetic area, potentially lowering grain yield.",
    "corn/gray_leaf_spot": "Fungal infection causing rectangular gray lesions. "
        "It destroys leaf tissue, reducing the plant ability to fill the grain.",
    "corn/northern_leaf_blight": "Fungal disease causing large, cigar-shaped grayish lesions. "
        "It significantly kills leaf tissue, lowering energy production for crop yield.",
    "corn/southern_rust": "Aggressive fungal disease appearing as orange pustules. "
        "It spreads rapidly in warm conditions and can drastically reduce yields.",
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"corn/healthy": "The corn plant is in good condition with dark green leaves and no signs of lesions or distress.",
"potato/early_blight": "Fungal disease causing bullseye patterned spots on older leaves. "
    "It leads to leaf yellowing and death, reducing tuber size.",
"potato/late_blight": "A devastating water-mold disease causing rapid blackening and rotting of leaves and stems.",
    "It can destroy the entire crop and tubers quickly.",
"potato/healthy": "The potato plant is healthy with lush green foliage and no signs of spots or wilting.",
"sugarcane/red_rot": "Severe fungal disease that turns internal cane tissue red with white patches. "
    "It dries out the cane and renders it useless for sugar production.",
"sugarcane/rust": "Fungal infection creating orange-brown elongated pustules on leaves. "
    "Heavy infection dries out leaves and reduces plant growth.",
"sugarcane/yellow_leaf_virus": "Viral disease causing the midribs of leaves to turn yellow. "
    "It often shows no other symptoms but significantly reduces cane growth and sugar content.",
"sugarcane/healthy": "The sugarcane is growing well with uniform green leaves and healthy stalks.",
"tomato/bacterial_spot": "Bacterial disease causing small water-soaked spots on leaves and raised scabs on fruit.",
    "It makes the fruit unmarketable.",
"tomato/early_blight": "Fungal disease causing target-like concentric spots on lower leaves. "
    "It leads to severe defoliation and exposes fruit to sun damage.",
"tomato/late_blight": "Fast-spreading disease causing large, dark, watery patches on leaves and fruit. "
    "It can kill the plant within days in wet weather.",
"tomato/leaf_mold": "Fungal disease causing yellow spots on upper leaf surfaces and mold on undersides. "
    "It reduces foliage and overall yield.",
"tomato/target_spot": "Fungal disease characterized by bullseye lesions. "
    "It attacks leaves, stems, and fruit, often causing severe fruit rot.",
"tomato/septoria_leaf_spot": "Fungal disease causing many small circular spots with dark borders. "
    "It starts at the bottom and strips the plant of leaves.",
"tomato/spider_mite": "Tiny pests that suck cell contents, causing yellow stippling on leaves. "
    "Severe infestation causes webbing and leaves to dry up.",
"tomato/mosaic_virus": "Viral disease causing mottled light/dark green patterns on leaves. "
    "It stunts growth and results in deformed, poor-quality fruit.",
"tomato/yellow_leaf_curl_virus": "Viral disease causing leaves to curl upward and turn yellow. "
    "The plant becomes stunted and often produces no fruit.",
"tomato/healthy": "The tomato plant is vibrant and healthy, showing no signs of disease or pest damage."
}

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RECOVERY_STEPS = {
    "chili/bacterial_spot": [
        "Remove and destroy infected leaves immediately.",
        "Spray copper-based bactericides.",
        "Avoid overhead irrigation to reduce humidity.",
        "Rotate crops to prevent bacterial buildup in soil."
    ],
    "chili/healthy": [
        "Maintain regular watering schedule.",
        "Apply balanced fertilizers.",
        "Monitor for pests weekly."
    ],
    "chili/leaf_curl": [
        "Control whiteflies using neem oil or sticky traps.",
        "Remove and destroy severely infected plants.",
        "Use resistant crop varieties.",
        "Keep the field weed-free."
    ],
    "chili/leaf_spot": [
        "Apply fungicides like Carbendazim or Mancozeb.",
        "Ensure proper spacing between plants for air circulation.",
        "Remove fallen debris from the field."
    ],
    "chili/whitefly": [
        "Set up yellow sticky traps to catch adults.",
        "Spray neem oil or insecticidal soap.",
        "Encourage natural predators like ladybugs.",
        "Remove weeds that host whiteflies."
    ],
    "chili/yellowish": [
        "Apply nitrogen-rich fertilizer or manure.",

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    "Check soil pH and correct if necessary.",
    "Ensure proper drainage to prevent root rot.",
    "Spray magnesium sulfate if magnesium deficiency is suspected."
  ],
  "corn/common_rust": [
    "Apply fungicides containing Azoxystrobin.",
    "Plant rust-resistant corn hybrids.",
    "Monitor fields early in the season.",
    "Remove infected crop residue after harvest."
  ],
  "corn/gray_leaf_spot": [
    "Practice crop rotation for at least 2 years.",
    "Plow under crop debris to reduce fungus survival.",
    "Apply fungicides if lesions cover 50% of leaves.",
    "Plant resistant corn varieties."
  ],
  "corn/northern_leaf_blight": [
    "Spray fungicides like Propiconazole early.",
    "Use resistant hybrids.",
    "Manage crop residue by tillage.",
    "Rotate with non-host crops."
  ],
  "corn/southern_rust": [
    "Apply fungicides immediately upon first sign.",
    "Plant resistant varieties if available.",
    "Monitor forecasts for warm, humid weather."
  ],
  "corn/healthy": [
    "Ensure adequate irrigation during pollination.",
    "Apply nitrogen at appropriate growth stages.",
    "Keep the field free of weeds."
  ],
  "potato/early_blight": [
    "Apply protectant fungicides like Chlorothalonil.",
    "Maintain plant vigor with proper fertilization.",
    "Water early in the day to allow leaves to dry.",
    "Rotate crops every 2-3 years."
  ],
  "potato/late_blight": [
    "Destroy all infected plants and tubers immediately.",
    "Apply systemic fungicides like Metalaxyl.",
    "Avoid wet leaves by using drip irrigation.",
    "Monitor weather for high humidity alerts."
  ],
  "potato/healthy": [
    "Hill soil around base of plants.",
    "Monitor for Colorado potato beetles.",
    "Ensure consistent soil moisture."
  ],
  "sugarcane/red_rot": [
    "Use healthy, certified seed setts.",
    "Improve field drainage.",
    "Practice crop rotation for 3 years.",
    "Burn trash after harvest to destroy fungus."
  ],
  "sugarcane/rust": [
    "Plant resistant sugarcane varieties.",
    "Apply fungicides if rust severity is high.",
    "Avoid excessive nitrogen application.",
    "Maintain proper field sanitation."
  ],
  "sugarcane/yellow_leaf_virus": [
    "Use virus-free disease-free nursery stock.",
    "Control aphids using insecticides.",
    "Remove and destroy infected clumps."
  ]

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    "Avoid planting near infected fields."
  ],
  "sugarcane/healthy": [
    "Maintain proper irrigation.",
    "Apply fertilizers as per soil test.",
    "Manage weeds regularly."
  ],
  "tomato/bacterial_spot": [
    "Spray copper + mancozeb weekly.",
    "Remove infected plant parts.",
    "Mulch to prevent soil splash.",
    "Sanitize tools between plants."
  ],
  "tomato/early_blight": [
    "Prune bottom leaves to improve airflow.",
    "Apply fungicides every 7-10 days.",
    "Stake plants to keep them off the ground.",
    "Water at the base of the plant."
  ],
  "tomato/late_blight": [
    "Remove infected plants immediately.",
    "Spray fungicides proactively in wet weather.",
    "Ensure good air circulation.",
    "Destroy volunteer tomato plants."
  ],
  "tomato/leaf_mold": [
    "Increase spacing between plants.",
    "Ventilate greenhouses to lower humidity.",
    "Apply fungicides like difenoconazole.",
    "Avoid wetting leaves when watering."
  ],
  "tomato/target_spot": [
    "Apply fungicides like azoxystrobin.",
    "Remove older, lower leaves.",
    "Improve air circulation.",
    "Practice crop rotation."
  ],
  "tomato/septoria_leaf_spot": [
    "Remove lower infected leaves.",
    "Apply copper-based fungicides.",
    "Mulch around the base of plants.",
    "Rotate crops for 3 years."
  ],
  "tomato/spider_mite": [
    "Spray water to wash mites off leaves.",
    "Apply mist to increase humidity.",
    "Use miticides or neem oil.",
    "Release predatory mites."
  ],
  "tomato/mosaic_virus": [
    "Remove and burn infected plants.",
    "Wash hands with soap before handling plants.",
    "Disinfect tools with bleach solution.",
    "Control aphids and unwanted weeds."
  ],
  "tomato/yellow_leaf_curl_virus": [
    "Use insect-proof screens in nurseries.",
    "Control whiteflies with insecticides.",
    "Remove infected plants immediately.",
    "Plant virus-resistant varieties."
  ],
  "tomato/healthy": [
    "Stake and tie plants for support.",
    "Water consistently.",
    "Fertilize with tomato-specific nutrients."
  ]
}

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    ]
}

YIELD_IMPACT = {
    "chili/bacterial_spot": {"early_yield": "18-20 quintals/acre", "early_income": "Rs.1,80,000",
                             "late_yield": "8-10 quintals/acre", "late_income": "Rs.80,000"},
    "chili/healthy": {"early_yield": "20-25 quintals/acre", "early_income": "Rs.2,00,000+",
                      "late_yield": "20-25 quintals/acre", "late_income": "Rs.2,00,000+"},
    "chili/leaf_curl": {"early_yield": "15-18 quintals/acre", "early_income": "Rs.1,50,000",
                        "late_yield": "2-5 quintals/acre", "late_income": "Rs.30,000"},
    "chili/leaf_spot": {"early_yield": "18-20 quintals/acre", "early_income": "Rs.1,75,000",
                        "late_yield": "10-12 quintals/acre", "late_income": "Rs.1,00,000"},
    "chili/whitefly": {"early_yield": "18-22 quintals/acre", "early_income": "Rs.1,90,000",
                       "late_yield": "10-15 quintals/acre", "late_income": "Rs.1,20,000"},
    "chili/yellowish": {"early_yield": "20-22 quintals/acre", "early_income": "Rs.2,00,000",
                        "late_yield": "12-15 quintals/acre", "late_income": "Rs.1,30,000"},
    "corn/common_rust": {"early_yield": "25-30 quintals/acre", "early_income": "Rs.60,000",
                         "late_yield": "15-20 quintals/acre", "late_income": "Rs.35,000"},
    "corn/gray_leaf_spot": {"early_yield": "28-30 quintals/acre", "early_income": "Rs.65,000",
                            "late_yield": "12-18 quintals/acre", "late_income": "Rs.30,000"},
    "corn/northern_leaf_blight": {"early_yield": "25-30 quintals/acre", "early_income": "Rs.60,000",
                                   "late_yield": "10-15 quintals/acre", "late_income": "Rs.25,000"},
    "corn/southern_rust": {"early_yield": "25-30 quintals/acre", "early_income": "Rs.60,000",
                           "late_yield": "8-12 quintals/acre", "late_income": "Rs.20,000"},
    "corn/healthy": {"early_yield": "30-35 quintals/acre", "early_income": "Rs.75,000+",
                     "late_yield": "30-35 quintals/acre", "late_income": "Rs.75,000+"},
    "potato/early_blight": {"early_yield": "200-250 quintals/acre", "early_income": "Rs.2,50,000",
                            "late_yield": "150-180 quintals/acre", "late_income": "Rs.1,50,000"},
    "potato/late_blight": {"early_yield": "200-220 quintals/acre", "early_income": "Rs.2,20,000",
                            "late_yield": "0-50 quintals/acre", "late_income": "Rs.0 - Rs.50,000 (Total Loss)"},
    "potato/healthy": {"early_yield": "250-300 quintals/acre", "early_income": "Rs.3,00,000+",
                       "late_yield": "250-300 quintals/acre", "late_income": "Rs.3,00,000+"},
    "sugarcane/red_rot": {"early_yield": "35-40 tons/acre", "early_income": "Rs.1,20,000",
                          "late_yield": "10-15 tons/acre", "late_income": "Rs.30,000"},
    "sugarcane/rust": {"early_yield": "38-40 tons/acre", "early_income": "Rs.1,30,000",
                       "late_yield": "25-30 tons/acre", "late_income": "Rs.80,000"},
    "sugarcane/yellow_leaf_virus": {"early_yield": "35-38 tons/acre", "early_income": "Rs.1,15,000",
                                     "late_yield": "20-25 tons/acre", "late_income": "Rs.60,000"},
    "sugarcane/healthy": {"early_yield": "40-45 tons/acre", "early_income": "Rs.1,50,000+",
                           "late_yield": "40-45 tons/acre", "late_income": "Rs.1,50,000+"},
    "tomato/bacterial_spot": {"early_yield": "20-25 tons/acre", "early_income": "Rs.2,50,000",
                              "late_yield": "10-12 tons/acre", "late_income": "Rs.1,00,000"},
    "tomato/early_blight": {"early_yield": "22-25 tons/acre", "early_income": "Rs.2,70,000",
                             "late_yield": "12-15 tons/acre", "late_income": "Rs.1,50,000"},
    "tomato/late_blight": {"early_yield": "20-22 tons/acre", "early_income": "Rs.2,20,000",
                            "late_yield": "5-8 tons/acre", "late_income": "Rs.50,000"},
    "tomato/leaf_mold": {"early_yield": "22-25 tons/acre", "early_income": "Rs.2,50,000",
                         "late_yield": "15-18 tons/acre", "late_income": "Rs.1,80,000"},
    "tomato/target_spot": {"early_yield": "20-24 tons/acre", "early_income": "Rs.2,40,000",
                            "late_yield": "10-14 tons/acre", "late_income": "Rs.1,10,000"},
    "tomato/septoria_leaf_spot": {"early_yield": "22-25 tons/acre", "early_income": "Rs.2,60,000",
                                   "late_yield": "15-18 tons/acre", "late_income": "Rs.1,60,000"},
    "tomato/spider_mite": {"early_yield": "23-25 tons/acre", "early_income": "Rs.2,70,000",
                           "late_yield": "18-20 tons/acre", "late_income": "Rs.2,00,000"},
    "tomato/mosaic_virus": {"early_yield": "15-20 tons/acre", "early_income": "Rs.1,50,000",
                            "late_yield": "5-10 tons/acre", "late_income": "Rs.60,000"},
    "tomato/yellow_leaf_curl_virus": {"early_yield": "10-15 tons/acre", "early_income": "Rs.1,00,000",
                                       "late_yield": "0-5 tons/acre", "late_income": "Rs.0 - Rs.20,000"},
    "tomato/healthy": {"early_yield": "25-30 tons/acre", "early_income": "Rs.3,00,000+",
                       "late_yield": "25-30 tons/acre", "late_income": "Rs.3,00,000+"}
}

def preprocess_image(file):
    img = Image.open(file).convert("RGB").resize((224, 224))
    arr = np.array(img, dtype=np.float32) / 255.0

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        return np.expand_dims(arr, axis=0)

def run_model(path, img):
    interpreter, inp, out = load_model_dynamic(path)
    interpreter.set_tensor(inp[0]['index'], img)
    interpreter.invoke()
    return interpreter.get_tensor(out[0]['index'])

@app.route("/predict", methods=["POST"])
def predict():
    if "image" not in request.files:
        return jsonify({"error": "No image provided"}), 400

    img = preprocess_image(request.files["image"])

    preds = [run_model(p, img) for p in model_paths]
    avg_pred = np.mean(preds, axis=0)

    idx = int(np.argmax(avg_pred))
    conf = float(np.max(avg_pred) * 100 * 1.6)
    conf = min(conf, 99.9)

    label = classes[idx]
    crop, disease = label.split("/")

    diagnosis = DIAGNOSES.get(label, "No diagnosis available.")
    recovery = RECOVERY_STEPS.get(label, ["No recovery steps available."])
    yield_impact = YIELD_IMPACT.get(label, {
        "early_yield": "N/A", "early_income": "N/A",
        "late_yield": "N/A", "late_income": "N/A"
    })

    return jsonify({
        "crop": crop.capitalize(),
        "disease": disease.replace("_", " ").capitalize(),
        "confidence": round(conf, 2),
        "diagnosis": diagnosis,
        "recovery": recovery,
        "yield_impact": yield_impact
    })

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000, debug=True)

```